

What is backpropagation(Neural network) ?

Select one:

- ☒ a. It is the transfer of error back through the network using a set of recurrent connections.
- ☐ b. It is the transfer of outputs back from the hidden layer to the input layer using a set of recurrent connections.
- ☐ c. It is the transfer of error back through the network to adjust the inputs.
- ☐ d. It is the transfer of error back through the network to allow the weights to be adjusted.

Decision tree is an algorithm for which machine learning task?

Select one:

- ☐ a. clustering
- ☐ b. regression
- ☒ c. classification
- ☐ d. dimensionality reduction



44. What Is The Purpose of Aggregation (in Fuzzy Inference System)?

Select one:

- ☒ a. To gather all the different fuzzy set outputs and combine them into a single fuzzy set outputs
- ☐ b. To subtract all the output fuzzy set values from the input values
- ☐ c. To gather all the different fuzzy set outputs and average them out to get a single value
- ☐ d. To gather all the possible inputs and use the average to gain an output

Previous probabilities in Bayes Theorem that are changed with help of new available information are classified as:

Select one:

- ☐ a. interior probabilities
- ☐ b. dependent probabilities
- ☒ c. posterior probabilities
- ☐ d. independent probabilities



31. The support of Fuzzy Set A is the set of all points x in X (is the universe of discourse) such that

Select one:

- ☐ a. Value of membership function of $x = 0$
- ☒ b. Value of membership function of $x > 0$
- ☐ c. Value of membership function of $x = 1$
- ☐ d. Value of membership function of $x = 0.5$

A 4-input neuron has weights 1, 2, 3 and 4. The transfer function is linear i.e. ($f(x)=2 \cdot x$). The inputs are 4, 10, 5 and 20 respectively. The output will be:

Select one:

☒ a. 238

☐ b. 123

☐ c. 76

☐ d. 119



We have trained a machine learning model to identify that whether the object in an image is a cat. Now we use 200 images to test its performance. Among 200 images, 170 contain cats and 30 do not contain cats.

The identification result of the model is that 140 images contain cats and 60 do not contain cats.

Given that TP (true positive) of the model = 137.

What is the recall of the model?

Select one:

- ☐ a. 82.5 %
- ☒ b. 80.5 %
- ☐ c. 96.4 %
- ☐ d. 75.7 %



If x is A then y is B else y is C . The output of the given fuzzy rule is

Select one:

- ☐ a. a fuzzy set
- ☐ b. a crisp set
- ☐ c. a membership function
- ☒ d. a fuzzy relation



Suppose that Ali can decide to go to work by one of three modes of transportation, car, bus, or train. Because of high traffic, if he decides to go by car, there is a 60 % chance he will be late. If he goes by bus, the probability of being late is 30%. The train is late with a probability of only 15 %.

One day Ali is late. Answer the following questions given that $p(\text{car})=6/12$, $p(\text{bus}) = 1/12$ $p(\text{train}) = 5/12$.

Hint: use $h_{\text{MAP}} = \arg \max P(h/D) = \arg \max P(D/h) P(h)$

A. Compute $p(\text{late}/\text{car}) * p(\text{car})$? 0.3

B. Compute $p(\text{late}/\text{train}) * p(\text{train})$? 0.062

C. Compute $p(\text{late}/\text{bus}) * p(\text{bus})$? 0.025

D. What mode of transportation did he use? car

Consider the training examples shown in the Table below for a binary classification problem.

Where a_1 , a_2 and a_3 are the attributes

Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	-
4	F	F	4.0	+
5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

A. What is the entropy of this collection of training examples? 0.9911

B. What is the information gains of a_1 relative to these training examples? 0.7616

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6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

A. What is the entropy of this collection of training examples

0.9911



Choose...

0.9911

0.9111

0.9711

0.9811

B. What is the information gains of a_1 relative to these train

0.7616



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5	F	T	7.0	-
6	F	T	3.0	-
7	F	F	8.0	-
8	T	F	7.0	+
9	F	T	5.0	-

A. What is the entropy of this collection of training examples 0.9911

B. What is the information gains of a_1 relative to these training examples? 0.7616

Choose...

0.7511

0.7616

0.7712

0.7451

Suppose a fuzzy set Young is defined as follows:

Young = {(10, 0.5), (20, 0.8), (30, 0.8), (40, 0.5), (50, 0.3)}

Then the crisp value of Young using Mean of maxima (MoM) defuzzification method is

Select one:

- ☐ a. 25
- ☒ b. 50
- ☐ c. 20
- ☐ d. 35



A regression between foot length (response variable in cm y) and height (explanatory variable in inches x) for 33 students resulted in the following regression equation:

$$y = 10.9 + 0.23x$$

What is the estimated average foot length for students who are 70 inches tall?

Select one:

- ☐ a. 28 cm
- ☐ b. 29 cm
- ☐ c. 30 cm
- ☒ d. 27 cm



1. Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$$

$$\text{and } B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$$

Find $A \cup B = ?$

Select one:

- ☐ a. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$
- ☒ b. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 0.2/e\}$
- ☐ c. $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$
- ☐ d. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$



Which of the following methods do we use to best fit the data in Logistic Regression?

Select one:

- ☐ a. Least Square Error
- ☐ b. Jaccard distance
- ☐ c. Both least square and maximum likelihood
- ☒ d. Maximum Likelihood



Which statement is wrong related to a decision tree?

Select one:

- ☐ a. A decision tree is a tree-like structure (a binary tree or a non-binary tree).
- ☐ b. Each leaf node stores a class.
- ☐ c. Each non-leaf node represents a test on a feature attribute.
- ☒ d. Each branch stores the entropy of attribute.

Consider two fuzzy subsets of the set X , $X = \{a, b, c, d, e\}$ referred to as A and B .

$A = \{1/a, 0.3/b, 0.2/c, 0.8/d, 0/e\}$

and $B = \{0.6/a, 0.9/b, 0.1/c, 0.3/d, 0.2/e\}$

Find $A' = ?$

Select one:

- ☐ a. $\{1/a, 0.9/b, 0.2/c, 0.8/d, 2/e\}$
- ☐ b. $\{0.4/a, 0.1/b, 0.9/c, 0.7/d, 0.8/e\}$
- ☒ c. $\{0/a, 0.7/b, 0.8/c, 0.2/d, 1/e\}$
- ☐ d. $\{0.6/a, 0.3/b, 0.1/c, 0.3/d, 0/e\}$



What is the name of this **task** that can be solved with machine learning: "In this type of **task**, the computer program is asked to predict a numerical value given some input."

Select one:

- ☐ a. Classification
- ☐ b. Generalization
- ☒ c. Regression
- ☐ d. Crossvalidation



The following data: (1, 1), (20, 400), (30, 800), (40, 1300) is regressed with least squares regression to $y = b + vx$. The value of w most nearly is

Select one:

- ☐ a. 28.956
- ☐ b. 27.480
- ☐ c. 40.000
- ☒ d. 32.625



Three companies A, B and C supply 25%, 35% and 40% of the notebooks to a school. Past experience shows that 5%, 4% and 2% of the notebooks produced by these companies are defective. If a notebook was found to be defective, what is the probability that the notebook was supplied by A?

Hint: Let D be the event that notebooks are defective then $P(A|D)$?

Select one:

- ☒ a. $\frac{25}{69}$
- ☐ b. $\frac{44}{69}$
- ☐ c. $\frac{13}{24}$
- ☐ d. $\frac{11}{24}$



Given a dataset of COVID-19 infection:

ID	Fever	Cough	Breathing issues	Infected
1	NO	NO	NO	NO
2	YES	YES	YES	YES
3	YES	YES	NO	NO
4	YES	NO	YES	YES
5	YES	YES	YES	YES
6	NO	YES	NO	NO
7	YES	NO	YES	YES
8	YES	NO	YES	YES
9	NO	YES	YES	YES
10	YES	YES	NO	YES
11	NO	YES	NO	NO
12	NO	YES	YES	NO
13	NO	YES	YES	NO
14	YES	YES	NO	NO

Using ID3 algorithm compute IG(S, Fever) (Information gain of fever)?

Hint: Final answer should contain exactly 2 digits after decimal point.(خاتین عشرين فقط)

Answer: 0.27